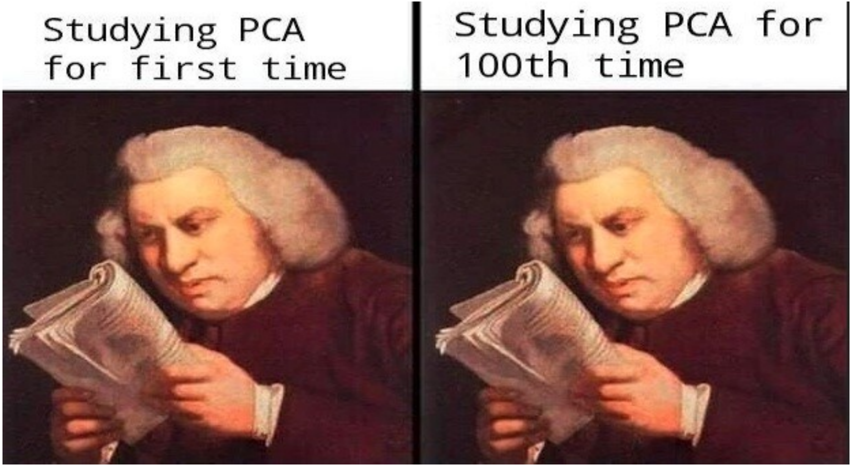
BIOS 794: Dimensions Reduction II

Alexander McLain

October 28, 2025

Outline

Introdoction



What is Principal Component Analysis (PCA)?

- PCA is a method to simplify high-dimensional data by finding new variables (called **principal components**) that summarize most of the variation in the data.
- Each principal component is a **weighted combination** of the original variables.
- These components are designed to capture patterns and relationships among the variables.

Example

We might have many related health indicators (e.g., blood pressure, BMI, cholesterol, glucose). PCA finds new variables that summarize these into a few uncorrelated "health factor" scores.

The Main Goal of PCA

- PCA finds directions (called **principal components**) in the data that explain the most variation.
- The first component captures the largest possible variance in the data.
- The second captures the next largest amount, and so on.
- Each new component must be **uncorrelated** with those before it.

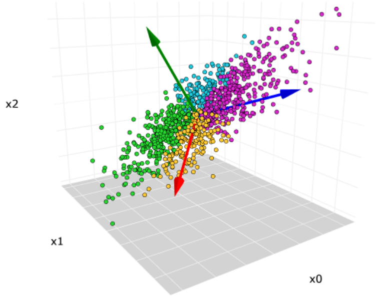


Figure: PCA finds the axes that explain the most variation in the data.

How PCA Works (Conceptually)

- Imagine plotting all observations on a scatterplot (e.g., BMI vs. cholesterol).
- PCA rotates the coordinate system to find the directions where points vary the most.
- These directions are found by solving an **eigenvalue problem** on the covariance (or correlation) matrix.
- The eigenvalues tell us how much variation each component explains.

No need to memorize the math!

Just remember: PCA creates new variables that are:

1. Uncorrelated,
2. Ordered by how much total variation they explain.

Understanding PCA Output

- **Eigenvalues** (d_i): How much variance each principal component explains.
- **Loadings**: The weights (or coefficients) that define each principal component in terms of the original variables.
- **Scores**: The value of each principal component for every observation.

Interpretation Tip

Large loadings (positive or negative) show which variables are most influential in that component.

How Many Components to Keep?

- We usually keep the smallest number of components that explain most of the total variation.
- **Cumulative variance explained:**

$$\frac{d_1 + d_2 + \ldots + d_r}{\sum_i d_i}$$

- **Scree plot:** A plot of eigenvalues (look for the "elbow").
- Common rules:
  - Keep components explaining 80–95% of the variance.
  - *Kaiser's Rule:* Keep components with eigenvalues > 1 (if using correlations).

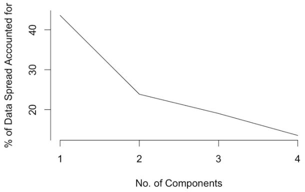


Figure: Example scree plot.

Key Takeaways

- PCA simplifies complex data by creating new uncorrelated variables.
- The first few components often explain most of the important variation.
- Use PCA when:
  - Variables are correlated.
  - You want to reduce dimensionality before modeling (e.g., regression, clustering).
- PCA depends on the scale of the variables — use standardized data when units differ.

In public health...

PCA is often used to create:

- Socioeconomic indices,
- Diet quality scores,
- Composite exposure metrics.

Advantages of PCA

1. Efficiency
2. Noise Reduction
3. Unsupervised
4. Global structure

Disadvantages of PCA

1. Linearity
2. Loss of Information
3. Sensitivity to Scaling
4. Not Robust to Outliers
5. Lack of Emphasis on Local Structure

Introduction to Factor Analysis

- Factor Analysis (FA) models correlations among observed variables using a smaller number of unobserved **latent factors**.
- Each observed variable is expressed as a linear combination of common factors plus unique (specific) error.
- The goal is to uncover underlying constructs that explain shared variability in the data.

Example:

- Observed variables: responses to depression-related survey questions.
- Latent factor: overall level of depressive symptomatology.

Model Formulation

Factor model

$$\mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \boldsymbol{\epsilon}$$

- $\mathbf{X}$: $p \times 1$ vector of observed variables
- $\mathbf{F}$: $m \times 1$ vector of latent factors ($m < p$)
- $\boldsymbol{\Lambda}$: $p \times m$ matrix of **factor loadings**
- ϵ: $p \times 1$ vector of unique (specific) errors

Covariance structure:

$$\Sigma = \Lambda\Lambda' + \Psi$$

where Ψ is a diagonal matrix of unique variances.

Interpretation

- Each factor explains correlations among a group of variables.
- The **factor loadings** show how strongly each variable relates to each latent factor.
- Factors can often be interpreted meaningfully (e.g., "psychological well-being," "socioeconomic status").
- Factor scores can be computed and used in regression or prediction models.

Note:

- Rotations (e.g., varimax, oblimin) are used to improve interpretability.

Factor Analysis vs. PCA

	PCA	**Factor Analysis**
Goal	Maximize explained variance	Model shared variance (covariance structure)
Latent variables	Linear combinations of observed variables	Underlying unobserved constructs
Error term	Not explicitly modeled	Unique variances modeled (Ψ)
Interpretability	Descriptive	Inferential / theoretical

Example in Public Health

- Suppose we have 10 survey items about:
  - Emotional symptoms
  - Sleep issues
  - Fatigue and energy
- Factor analysis might reveal:
  - Factor 1: Emotional well-being
  - Factor 2: Physical fatigue
- The factor scores can then be used as predictors of outcomes like depression diagnosis or productivity.

Introduction to ICA

- Independent Component Analysis (ICA) separates a multivariate signal into additive, statistically independent components.
- Originally developed for "blind source separation" — e.g., separating voices in an audio recording (the "cocktail party problem").
- Similar to FA, but assumes **statistical independence** rather than correlation structure.

Applications:

- Neuroimaging (fMRI, EEG) — identify independent brain activation patterns.
- Genomics — find independent biological signals.

ICA Model

$$\mathbf{X} = \mathbf{AS}$$

- $\mathbf{X}$: observed data (mixtures of signals)
- $\mathbf{S}$: independent components (latent sources)
- $\mathbf{A}$: mixing matrix (weights combining the sources)

Goal: Estimate the unmixing matrix $\mathbf{W} = \mathbf{A}^{-1}$, such that:

$$\mathbf{S} = \mathbf{WX}$$

and the elements of $\mathbf{S}$ are as independent as possible.

Comparison: FA, PCA, and ICA

	PCA	**FA / ICA**
Goal	Uncorrelated components	Independent components (ICA) or shared variance (FA)
Assumption	Orthogonal directions	Statistical independence (ICA) or covariance structure (FA)
Nature	Linear, deterministic	Linear, probabilistic (FA), or non-Gaussian (ICA)
Applications	Compression, visualization	Signal separation, latent constructs

ICA in Public Health

- **Neuroimaging:** Separate independent brain activation networks from fMRI data.
- **Environmental health:** Identify independent pollution sources (e.g., traffic, industry) from mixed sensor readings.
- **Genomics:** Extract independent expression patterns or pathways from high-dimensional omics data.

Key idea: While PCA finds directions of maximum variance, ICA finds statistically independent signals hidden within that variance.

Summary

- **Factor Analysis:** models shared variance using latent constructs — useful for understanding underlying dimensions.
- **ICA:** separates signals into independent sources — useful for uncovering hidden, non-Gaussian patterns.
- Both can reduce dimensionality, but serve distinct purposes:
  - FA: interpretability and theory-driven latent traits.
  - ICA: signal separation and independence.

What is MDS?

- Multidimensional Scaling (MDS) helps visualize how similar or dissimilar observations are.
- It finds a map (often 2D or 3D) so that points close in the original data remain close on the plot.
- Think of it as: "How can we place these points on a map so their distances reflect their similarity?"
- We'll focus on **classical (metric) MDS**, which preserves actual distances rather than just their rank order.

How MDS Works (Conceptually)

1. Start with a matrix of pairwise distances between subjects (e.g., Euclidean distances).
2. MDS finds coordinates for each subject in 2D or 3D so that the distances between them match the original distances as closely as possible.
3. This involves eigenvalue decomposition (behind the scenes).

Key idea

Similar subjects appear close together in the map; dissimilar subjects appear far apart.

Why Use MDS?

- Classical MDS is a global method because it seeks to preserve all pairwise distances between points, maintaining the overall structure of the data rather than just local neighborhoods.
- **Visualization:** Great for exploring structure in survey, genetic, or behavioral data.
- **Distance-based:** Works with any dissimilarity measure, not just Euclidean.
- **Invertable:** Can be used on new subjects.

Example

Visualizing similarity between states based on health indicators (obesity, smoking, physical activity, etc.).

What is t-SNE?

- **t-Distributed Stochastic Neighbor Embedding (t-SNE)** is a popular tool for visualizing high-dimensional data.
- It focuses on preserving **local structure**—keeping nearby points close.
- Often used to reveal clusters or patterns in complex biological, clinical, or social datasets.

How t-SNE Works (Conceptually)

1. Computes how similar points are in the original space (based on distance).
2. Tries to place points in 2D so that nearby points stay close, and faraway points stay apart.
3. Uses probabilities instead of raw distances—so it's good at capturing local neighborhoods.
4. The "t" in t-SNE comes from the t-distribution, which prevents crowding in 2D.

Tuning tip

Perplexity controls how much local vs. global structure is emphasized.

t-SNE in Practice

- Great for finding hidden clusters in patient or gene expression data.
- Each run may look slightly different—results are stochastic.
- Does not preserve global distances—don't interpret absolute distances between clusters.

Public health example

Visualizing patient subgroups based on behavioral and biomarker profiles.

UMAP

- **Uniform Manifold Approximation and Projection (UMAP)** is similar to t-SNE but faster and more flexible.
- Balances local and global structure—shows fine details without losing big-picture relationships.
- Works well for clustering, visualization, and feature reduction before modeling.

Advantages

- Faster than t-SNE.
- Better at showing both clusters and their relationships.
- Works with different distance measures.

PaCMAP

- **PaCMAP (Parameter-free Conformal Mapping)** is a newer technique.
- It preserves both local and global relationships without much tuning.
- Uses ideas from conformal prediction to improve reliability and reduce parameter dependence.

Key benefit

Consistent structure across runs—less sensitive to hyperparameters than t-SNE or UMAP.

Using Dimension Reduction in Regression

- You can use dimension-reduced variables (e.g., from PCA, MDS, UMAP) as predictors in a model.
- For PCA this is straightforward (**Principal Component Regression**).
- For nonlinear methods (e.g., t-SNE, UMAP), interpreting coefficients or predicting new subjects is more challenging.
- These methods are often used for **exploration** rather than direct inference.

Summary and Takeaways

- Dimension reduction simplifies complex datasets and reveals hidden structure.
- **Linear methods** (e.g., PCA, MDS) preserve global patterns.
- **Nonlinear methods** (e.g., Isomap, t-SNE, UMAP) preserve local neighborhoods.
- The best method depends on whether your goal is to visualize, cluster, or model.
- Always interpret reduced dimensions cautiously—they may not have direct meaning.